

Quiz 1

ECON 300: Labour Economics

Instructor: Muhebullah Karimzada

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Part A: Multiple Choice ($5 \times 1 = 5$ points)

Choose the best answer for each question.

1. Which of the following individuals is counted as **not in the labour force**?
 - (a) A worker who searched for a job last week
 - (b) A worker on temporary layoff
 - (c) A discouraged worker who stopped searching
 - (d) A worker who starts a new job next week
2. The employment-to-population ratio measures:
 - (a) Employment divided by the labour force
 - (b) Employment divided by the working-age population
 - (c) The fraction of unemployed workers
 - (d) The fraction of discouraged workers
3. In the labour–leisure model, an increase in non-labour income:
 - (a) Rotates the budget constraint
 - (b) Has no effect on labour supply
 - (c) Shifts the budget constraint upward in a parallel way
 - (d) Makes leisure cheaper relative to consumption
4. A backward-bending labour supply curve arises when:
 - (a) The substitution effect dominates at all wages
 - (b) Leisure is an inferior good
 - (c) The income effect dominates at high wages
 - (d) Non-labour income is zero

5. Which of the following policies is most likely to **reduce** measured labour force participation?
- (a) A rise in wages
 - (b) An expansion of unemployment insurance
 - (c) Improved job matching technology
 - (d) Lower payroll taxes

Problem 1: Labour Force Measurement and Discouragement (7.5 points)

A city has a working-age population (15+) of 900,000 people. The initial breakdown is:

- Employed: 585,000
 - Unemployed: 65,000
 - Not in the labour force: 250,000
- (a) (2.5 points) Calculate the **labour force**, the **labour force participation rate**, and the **unemployment rate**.
- (b) (2.5 points) Calculate the **employment-to-population ratio** and explain how it differs conceptually from the LFPR.
- (c) (2.5 points) Suppose an economic slowdown causes:
- 15,000 employed workers to lose their jobs, and
 - 10,000 unemployed workers to stop searching for work.

Recalculate the unemployment rate and explain why changes in discouragement complicate the interpretation of labour market conditions.

Problem 2: Labour Supply and Minimum Wage (7.5 points)

A worker's weekly budget constraint is:

$$C = 500 + 30h$$

and preferences are given by:

$$U = C - h^2$$

The labour market is described by:

$$L^D = 150 - 3w, \quad L^S = 30 + 2w$$

- (a) (2.5 points) Find the worker's **optimal hours of work**.
- (b) (2.5 points) Find the **competitive equilibrium wage and employment**.
- (c) (2.5 points) A minimum wage of $w = 25$ is introduced. Determine whether it is binding and calculate the resulting level of **unemployment**.

End of Quiz